④ Tekrar (3,5)'e göre ötele (Ters Öteleme)

$$\begin{bmatrix} -1 & -1 & 1 \\ 1 & 0 & 1 \\ 0 & -1 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 5 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 4 & 1 \\ 4 & 5 & 1 \\ 3 & 4 & 1 \end{bmatrix} \begin{matrix} A''' \\ B''' \\ C''' \end{matrix}$$

III. Kısım

⑤ $x \to -2$, $y \to 1$ ötele

$$\begin{bmatrix} 4 & 4 & 1 \\ 4 & 5 & 1 \\ 3 & 4 & 1 \end{bmatrix} \cdot \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -2 & 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 5 & 1 \\ 2 & 6 & 1 \\ 1 & 5 & 1 \end{bmatrix} \begin{matrix} A^4 \\ B^4 \\ C^4 \end{matrix}$$

⑥ 2 kat ölçekle (düzgün)

$$\begin{bmatrix} 2 & 5 & 1 \\ 2 & 6 & 1 \\ 1 & 5 & 1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 10 & 1 \\ 4 & 12 & 1 \\ 2 & 10 & 1 \end{bmatrix} \begin{matrix} A^5 \\ B^5 \\ C^5 \end{matrix}$$

Kübik Spline

Soru ⇒ P_1 [0 0]
P_2 [1 2]
P_3 [3 2]

Relax sınır koşullarını kullanarak bu noktalardan geçen kübik spline çiziniz

$(P_1\ P_2)$	$(P_2\ P_3)$
P_1 ✓	P_2 ✓
P_2 ✓	P_3 ✓
P_1' ×	P_2' ×
P_2' ×	P_3' ×
t_2 ✓	t_3 ✓

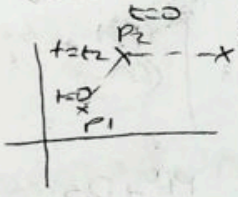
$t_2 = \sqrt{(1-0)^2 + (2-0)^2} = \sqrt{5}$

$t_3 = \sqrt{(3-1)^2 + (2-2)^2} = 2$

2. dereceden sürekli 2. dereceden türevleri, birbirine eşit. $P''(t_2) = \Theta''(0)$

$(P_1P_2) \sim (P_2P_3)$

$P''(t_2) = P''(0)$



$P(t) = B_1 + B_2 t + B_3 t^2 + B_4 t^3$

$P'(t) = B_2 + 2B_3 t + 3B_4 t^2$

$P''(t) = 2B_3 + 6B_4 t$

$P''(t_2) = 2B_3 + 6B_4 t_2 \quad \Big\rangle \quad \underbrace{2B_3 + 6B_4 t_2}_{P_1, P_2} = \underbrace{2B_3}_{P_2, P_3}$ ✓

$\Theta''(0) = 2B_3$

$$2\left[\frac{3(P_2-P_1)}{t_2^2} - \frac{2P_1'}{t_2} - \frac{P_2'}{t_2}\right] + 6\left[\frac{2(P_1-P_2)}{t_2^3} + \frac{P_1'}{t_2^2} + \frac{P_2'}{t_2^2}\right] = 2\left[\frac{3(P_3-P_2)}{t_3^2} - \frac{2P_2'}{t_3^2} - \frac{P_3'}{t_3^2}\right]$$

$$t_3 \cdot P_1' + 2(t_2+t_3) \cdot P_2' + t_2 P_3' = \frac{3}{t_2 t_3}\left[t_2^2(P_3-P_2) + t_3^2(P_2-P_1)\right] \to 2.\text{satır}$$

P_1 için $P''(0) = 0$ → başlangıç noktası için

$2B_3 = 0 \Rightarrow B_3 = 0$

$$P_1' + \frac{1}{2}P_2' = \frac{3}{2}\frac{(P_2-P_1)}{t_2} \to 1.\text{satır}$$

P_3 için $P''(t_3) = 0 \Rightarrow$ bitiş noktası için

$2B_3 + 6B_4 t_3 = 0$

$$2P_{n-1}' + 4P_n' = \frac{6}{t_n}(P_n - P_{n-1}) \to 3.\text{satır} \qquad 2P_2' + 4P_3' = \frac{6}{t_3}(P_3 - P_2)$$

$$\begin{bmatrix} 1 & 1/2 & 0 \\ t_3 & 2(t_2+t_3) & t_3 \\ 0 & 2 & 4 \end{bmatrix} \begin{bmatrix} P_1' \\ P_2' \\ P_3' \end{bmatrix} = \begin{bmatrix} 3/2 \frac{(P_2-P_1)}{t_2} \\ 3/t_2 t_3 [t_2^2 (P_3-P_2) + t_3^2 (P_2-P_1)] \\ 6/t_3 [(P_3-P_2)] \end{bmatrix}$$

$$\begin{bmatrix} 1 & 1/2 & 0 \\ 2 & 8{,}472 & 2{,}236 \\ 0 & 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} P_1' \\ P_2' \\ P_3' \end{bmatrix} = \begin{bmatrix} 3/2 \cdot \frac{[1\ 2]}{t_2} \\ 3/t_2 t_3 [2[2\ 0] + 2\sqrt{2}[1\ 2]] \\ 6/t_3 [2\ 0] \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0{,}5 & 0 \\ 2 & 8{,}472 & 2{,}236 \\ 0 & 2 & 4 \end{bmatrix} \cdot \begin{bmatrix} P_1' \\ P_2' \\ P_3' \end{bmatrix} = \begin{bmatrix} [0{,}670 & 1{,}341] \\ [9{,}391 & 5{,}366] \\ [6 & 0] \end{bmatrix}$$

2,0

$P(1\ 2)$

$0{,}670[5[2,0] + 4[1\ 2]]$

$P_1' + 0{,}5P_2' = [0{,}670\ \ 1{,}341]$

$2P_1' + 8{,}472P_2' + 2{,}236P_3' = [9{,}391\ \ 5{,}366]$

$2P_2' + 4P_3' = [6\ \ 0]$

$\rightarrow P_1' + 0{,}5P_2' = [0{,}670\ \ 1{,}341]$

2. derklem / 2 $\Rightarrow$ $P_1 + 4{,}236P_2' + 1{,}118P_3' = [4{,}695\ \ 2{,}683]$

3. derklem / 2 $\rightarrow$ $3{,}736P_2' + 1{,}118P_3' = [4{,}025\ \ 1{,}342]$

2. derklem / 3,736 $\Rightarrow$ $P_2' + 0{,}3P_3' = [1{,}077\ \ 0{,}359]$

$P_2' + 2P_3' = [3\ \ 0]$

$1{,}7P_3' = [1{,}923\ \ -0{,}359]$

$\boxed{P_3' = [1{,}131\ \ -0{,}211]}$

$P_2' + 2[1{,}131\ \ -0{,}211] = [3\ \ 0]$

$\boxed{P_2' = [0{,}738\ \ 0{,}422]}$

$P_1' + 0{,}5[0{,}738\ \ 0{,}422] = [0{,}670\ \ 1{,}341]$

$\boxed{P_1' = [0{,}301\ \ 1{,}131]}$

$|P_1P_2|$

$P(t) = B_1 + B_2 t + B_3 t^2 + B_4 t^3 \qquad 0 \le t \le 2{,}236$

$= P_1 + P_1' t + B_3 t^2 + B_4 t^3$

$= [0\ \ 0] + [0{,}301\ \ 1{,}131] t + [0\ \ 0] t^2 + [0{,}029\ \ -0{,}042] t^3$

$\frac{t}{t_2} = \frac{1}{3} = \frac{\sqrt{5}}{3} = P(0{,}745) = [0{,}2336\ \ 0{,}8227)$

$|P_2P_3|$

$P(t) \quad B_1 + B_2 t + B_3 t^2 + B_4 t^3 \qquad 0 \le t \le 2$

$B_2 = P_2'$

$= [1\ 2] + [0{,}7384\ \ 0{,}4226] t + [0{,}185535\ \ -0{,}31685] t^2 + [-0{,}0253\ \ 0{,}0283] t^3$

$\frac{t}{t_3} = \frac{1}{3} = P(2/3) = [1{,}573\ \ 2{,}1578]$